

Hardik Choudhary

hardikchoudhary3101@gmail.com | (412) 670-5587 | Pittsburgh, PA
[linkedin.com/in/hardik-choudhary-38524324a](https://www.linkedin.com/in/hardik-choudhary-38524324a) | Portfolio: hardik3101.github.io/portfolio

EDUCATION

Carnegie Mellon University <i>Master of Science in Mechanical Engineering – Research</i> Coursework: Modern Control Theory, Nonlinear Control, Introduction to Deep Learning, Machine Learning and AI for Engineers	Pittsburgh, PA May 2027
Sardar Vallabhbhai National Institute of Technology (SVNIT) <i>Bachelor of Technology in Mechanical Engineering GPA: 8.03/10.0</i>	Surat, India May 2024

SKILLS

Programming and Machine Learning: Python, C++, MATLAB, Simulink, PyTorch, NumPy, pandas
Robotics and Simulation: MuJoCo, Drake, Isaac Sim, Pinocchio, SOFA, Webots, ROS
Control and Planning: Trajectory optimization, graph search (INSAT, A*), LQR, adaptive control, PID, state observers, EKF SLAM
Hardware: Arduino, ESP32, NVIDIA Jetson Nano, Raspberry Pi, Quanser QUBE-Servo 2
Mechanical Design: SolidWorks (PDM/PLM), ANSYS (FEA), CFD, GD&T, design optimization, CNC machining, 3D printing

RESEARCH EXPERIENCE

Carnegie Mellon University <i>Graduate Student Researcher, On-Orbit Manipulation (Advisors: Howie Choset, Matt Travers)</i>	Pittsburgh, PA Aug 2025 – Present
- Extend the INSAT planner (interleaved graph search and trajectory optimization) to tumbling satellites with a space-time visibility graph, raising rendezvous success from 80% to 99% over 200 randomized trials - Develop grasp, base, and arm controllers that stop a satellite's spin within actuation limits in Drake, MuJoCo, and Isaac Sim - Analyze grasps for Snaker, a snake robot that wraps a satellite's engine nozzle in a helical cage, showing it tolerates up to 5x more misalignment than a parallel-jaw gripper and resists spin about every axis - Design a tendon-driven Snaker that one base motor coils into a helix, replacing per-joint motors because power and heat dissipation are limited in orbit; build an early 3D-printed prototype	
Indian Institute of Technology Gandhinagar <i>Project Assistant</i>	Gandhinagar, India Jan 2025 – Jun 2025
Designed a high-dexterity prosthetic foot and built a test bed to evaluate it under repeatable loading	

ACADEMIC PROJECTS

Deep Learning for Engine Remaining-Useful-Life Prediction Introduction to Deep Learning	Jan 2026 – May 2026
Benchmarked LSTM, Transformer, and temporal convolutional networks in PyTorch on NASA C-MAPSS turbofan data with 3 seeds each; the Transformer led at 13.66 ± 0.47 cycles RMSE	
Engine-Powered Drone SVNIT	Jan 2024 – Jun 2024
- Designed a single-engine drone for long flights in remote areas, lighter and cheaper than one carrying large battery packs, with a contact-free magnetic-gear drive to all four propellers - Controlled each propeller's speed by its gear gap through a servo-driven linkage; mapped speed against gap with an encoder and Arduino (up to 75% of the ideal 6,000 RPM) for feedback control that improved flight stability	
RAKSH Autonomous Agriculture Robot Team Lead, SVNIT	Feb 2023 – Dec 2023
- Developed row-following navigation for polyhouses with sensor fusion, VSLAM, and vision-based crop-row detection, running ROS and Python across Jetson Nano, Raspberry Pi, and ESP32 boards - Designed a slide-in 10-chemical spraying module, each container with a servo-driven valve on a shared line, so the robot sprays only the plants its machine-learning model flags as diseased - Designed a lead-screw linear actuator that lifts the camera to leaf height and implemented a rocker-bogie suspension; led a team of 5 to the Rs 1,00,000 Proof of Concept prize at Robofest-Gujarat 3.0	
Formula Student Electric Car Chassis Head, Team Phoenix Racing, SVNIT	Nov 2021 – Jan 2023
- Led a 30-member team through chassis design and fabrication; cut frame mass with ANSYS FEA within stiffness targets - Calculated suspension hardpoints and developed the aero package; won Concept Goals Management, Formula Bharat 2023	

PROFESSIONAL EXPERIENCE

Carnegie Mellon University , Pittsburgh, PA <i>Course Assistant, Machine Learning and AI for Engineers</i>	Aug 2026 – Present
<i>Course Assistant, AI Agents for Engineers</i>	Jan 2026 – May 2026
<i>Course Assistant, Applied Finite Element Analysis</i>	Aug 2025 – Dec 2025
Banaras Locomotive Works , Varanasi, India <i>Intern, locomotive assembly and quality control</i>	Jun 2023 – Jul 2023